

Dissertation Thesis

titled

“Evaluating the status of the globally threatened White Browed Bushchat *Saxicola macrorhynchus* in the western Thar landscape of Rajasthan.”

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by

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For the partial fulfillment of

**M.Sc. Biodiversity, Wildlife Conservation and Management
Semester IV**

Under the supervision of

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Bharatiya Vidya Bhavan's College

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University of Mumbai

(2023-2024)

Certificate

Date: 20th April 2024

Place: Mumbai

This is to certify that the thesis entitled “**Evaluating the status of the globally threatened White Browed Bushchat *Saxicola macrorhynchus* in the Thar Desert landscape, Rajasthan**” comprises original research work carried out by **Mr. Mihir Jadhav** (MSCBIOD 204), under my supervision from December 2023 to April 2024. No part of this thesis has been submitted in part or full to any other University for the award of any degree and it fulfils all the requirements laid down by the University of Mumbai.

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Declaration

I, Mihir Jadhav, hereby declare that, this thesis, titled “**Evaluating the status of the globally threatened White Browed Bushchat *Saxicola macrorhynchus* in the western Thar landscape of Rajasthan.**” is a presentation of my original research work, submitted to Bharatiya Vidya Bhavan, Bhavan’s College, Andheri (W) for the Partial Fulfilment of the Master’s Degree in Biodiversity, Wildlife Management, and Conservation. Wherever contributions of others are involved, every attempt has been made to indicate this clearly, with due reference to the literature and acknowledgment of collaborative research and discussions.

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In my capacity as the supervisor of the candidate’s thesis, I certify that the above statements are true to the best of my knowledge.

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Executive Summary

The vast expanses of the Thar, encompass one of the harshest regions in India. These also include one of the finest savanna grassland systems found in the country. The White-browed bushchat represents an infrequent passerine that inhabits this landscape. The species currently belongs to the category of “Vulnerable” in the IUCN red data repository. This can be attributed to large-scale land use land cover conversions occurring in the landscape. The dramatic livelihood shift from semi-pastoralism to agriculture has had its due impact on the biodiversity of the region. The population status of the bushchat is declining due to these large-scale habitat modifications. This study aims to explore the population status of the White-browed bushchat, associated with the landscape it inhabits. Following were the outcomes of this study:

1. Occupancy of the bushchat certainly showed steep changes in different landcover types; Restored grasslands (RG), Croplands (CL), and Rangelands (RL). The highest occupancy was found in grasslands followed by croplands and then by rangelands. Interestingly, the observations where the bushchat was found in croplands were always found on the fringes of the agricultural area, where small bushes of Chag (*Crotolaria burhia*), and grass patches of Sewan (*Lasiurus scindicus*), Ganthia (*Dactyloctenium aegyptium*), were found. This could also be attributed to the insect diversity found in croplands.
2. The density of the bushchat was found to be 2.13 individuals/kilometer. This was supplemented by significantly low encounter rates of the species. This portrays the infrequent status of this passerine. The overall abundance of this species for the study area was found to be 41 individuals.
3. The study showcased that the White-browed bushchat is heavily associated with grassland habitats and land use changes did have a significant impact on their occupancy probabilities. Intensively grazed rangelands showed the least occupancy probability for the bushchat.

The findings underscore the urgent need for concerted action to mitigate the impacts of land use changes and ensure the long-term survival of vulnerable species like the white-browed bushchat in their natural habitats.

1. Introduction

1.1. An introduction to Open Natural Ecosystems (ONEs).

Non-forested Open Natural Ecosystems (ONEs) constitute about two-thirds of the world's terrestrial habitat. These ecosystems play a vital role in sustaining rich biodiversity and supporting the livelihoods of numerous pastoral communities. However, these ecosystems face significant peril primarily due to land conversion (Madhusudan, M. D., & Vanak, A. T. 2022). Historically, these open natural ecosystems predominantly accommodated (semi) nomadic pastoral communities, utilizing grassland-savannah environments for extensive livestock grazing. Agricultural activities were minimal and primarily focused on the seasonal cultivation of resilient crops capable of withstanding the sparse and unpredictable rainfall patterns. This historically prevalent low-impact human presence in many arid zones facilitated adaptation among certain wildlife species to the altered habitat conditions, and in some cases, led to advantageous outcomes for local biodiversity (Safriel et al., 2005). The technological advancements in irrigation facilities for the last few decades have facilitated crop-based agriculture and are further incentivized by open market policies (Safriel et al., 2005). Such extensive agricultural practices homogenize habitats, ultimately leading to the loss of biodiversity (Benton et al., 2003; Donald, 2004).

The major chunk of the problem lies in the ambiguity of defining these open natural ecosystems. While it is relatively straightforward to delineate forested habitats, defining Non-forested open natural ecosystems (ONEs) encompasses a wide spectrum of landscapes. These ecosystems can encompass sandy deserts, rocky outcrops, and debris-strewn fields, extending to open grasslands interspersed with shrubs and sporadic trees found within tropical savannas (Ratnam et al., 2011). About 14,280 square kilometers of the Non-forested open natural ecosystems (ONEs) identified in mapping efforts are situated within India's Protected Area (PA) network. This area constitutes less than 5% of the total expanse of ONEs across India. Notably, five specific PAs, including Desert National Park, Kachch Desert Wildlife Sanctuary, Nagarjunsagar Srisailem Tiger Reserve, Kailadevi Wildlife Sanctuary, and Kaimur Wildlife Sanctuary, collectively cover 40% (5,653 square kilometers) of this area. Across India, ONEs span

approximately 319,675 square kilometers, accounting for roughly 10% of the country's total land cover (Madhusudan, M. D., & Vanak, A. T. 2022).

1.2. Gradient of land-use change in Rajasthan.

From 1880 to 2010, the human population in India has grown by sixfold. Population growth concurrently brought about significant Land Use Land Cover changes (LULC) in the Indian subcontinent. A substantial reduction of 23 million hectares in forest cover has been observed, with the most significant deforestation occurring predominantly during the period of British colonial rule. In contrast to the large-scale deforestation, cropland expanded from 92 million ha to a massive 140.1 million ha during 1880-2010 (H. Tian, K. Banger, & T. Bao 2014).

Rajasthan occupies 10% of India's geographical area and is currently the largest state. The population of this state shows a growth rate greater than the national average. According to the (2011 census), the growth rate of Rajasthan between 2010 and 2011 was 21.44 percent as opposed to the national average of 17.64 percent for India. This has indeed formed a precedent for the rising challenges of food security, fodder security, and health care security.

Before the advent of the green revolution and the introduction of irrigation infrastructure in the desert region, the traditional livelihood system in the Thar Desert was semi-pastoral. This means that people relied partially on animal husbandry for their sustenance. During the summer months, particularly in years with favorable rainfall, crops such as pearl millet, sesame, and pulses like moth and mung beans were cultivated (Varghese 2016). However, with the construction of the Indira Gandhi Canal and the subsequent increase in irrigation facilities, there was a notable shift in agricultural practices. Traditional Kharif crops were gradually replaced by cluster beans, fodder crops, groundnuts, and cotton. In the winter season, staple crops like wheat, mustard, chickpeas, and berseem (a type of rabi fodder) became predominant. Livestock composition has also shifted to small ruminants like sheep and goats. Livestock grazing constitutes immense migrations in search of good grazing stocks. Such migrations and large-scale displacement can be attributed to acute shortages of grazing lands. This in

turn is a result of hypertensive grazing pressures brought about by an increase in livestock populations (Varghese 2016)

1.3. Avian responses to the Land Use Change in landscapes.

The characteristic of being sensitive to small-scale changes in the habitat, and the ability to respond to the habitat changes, that can be perceived, make bird communities a suitable model for assessing the scale of impact brought about by changes in land use and land cover (Newbold 2015).

The diversity of bird species within a given ecosystem further underscores their importance as indicators of habitat quality and ecosystem health. Bird communities typically comprise a variety of species with different ecological niches and habitat preferences. Monitoring changes in the composition and diversity of bird communities in response to land cover changes provides valuable insights into the overall resilience of ecosystems. Additionally, understanding the interactions between birds and their habitats is essential for predicting how changes in land cover may affect biodiversity at broader scales.

Studying avian responses to land cover changes offers valuable insights into the ecological consequences of human activities. By focusing on birds as indicators of environmental change, researchers can assess the scale of impact and develop strategies to preserve biodiversity and ecosystem resilience. Effective conservation and land management practices depend on understanding how birds interact with their habitats and respond to changes in land cover over time.

1.4. Navigating through the dramatic shift in the Thar desert landscape.

The Thar extends from the western fringes of the Aravalis to the Indus in northwestern India. Subjected to excessive stress due to frequent draughts and scanty rainfall (occurring once every two to three years) (S. Poonia and A.S. Rao 2013). The 1,96,150

sq km area of which 70% falls in Rajasthan, encompasses the **largest protected area in the region**. Desert National Park, is a 3000 sq km region designated as an Important Bird Area (IBA) by Birdlife International (BI). The grasslands within Jaisalmer district of the Thar Desert hold immense significance for birdlife. Serving as the eastern boundary for numerous arid zone species and the western boundary for several semi-arid species of the Indian subcontinent, this region plays a pivotal role in avian biodiversity. Additionally, it serves as a crucial winter habitat for numerous Palearctic migratory birds and intersects with two major migratory flyways, the Central Asian Flyway and the West Asian East African Flyway. Notably, these grasslands represent the last stronghold for the critically endangered Great Indian Bustard, alongside providing refuge for various other threatened and range-restricted bird species.

The Thar Desert also holds the distinction of being the most densely populated desert globally, sustaining an approximate population density of 83 individuals per square kilometer. This demographic density has subjected the region to considerable biotic stress, a situation exacerbated in recent years by escalating human numbers (Islam & Rahmani 2011). Since the 1980s, there has been a notable shift towards a more settled lifestyle, marked by the expansion of agricultural activities across these expansive rangelands. This transition has coincided with a significant increase in both human and livestock populations, placing heightened pressure on land resources that were previously abundant. The conversion of rangelands into croplands has led to a reduction in available grazing areas, intensifying the grazing pressure on the remaining grasslands in the region. This dramatic growth in human and livestock populations has resulted in an unprecedented increase in the demand for resources to cater to modern livelihood and economic activities, adversely affecting biodiversity (Uddin 2021).

The advent of the Indira Gandhi Canal in the region has brought about a significant change in both the socioeconomic, and ecology of the region. This has brought about newer species from surrounding landscapes colonizing this previously inhospitable landscape (Rahmani & Soni, 1997).

1.5. Occupancy framework to assess the effects of land-use change on rare and threatened species.

In the ecosystem, species form the fundamental units of energy transfer and matter exchange. The dynamics of an ecosystem are often driven by species, and each species has a different scale of effect on ecosystem processes (Estes et al., 2011, Noon et al., 2012). Hence, understanding the patterns, status, and trends of these species is cardinal in drawing the efficiency of conservation practices. This being established, our study aims to explore the static occupancy of the White-browed Bushchat, which is based on two stochastic processes that determine the “detection” of the target species at a “site”.

One of the most congruous macro-ecological patterns has been the positive correlation between occupancy (proportion of sites occupied by a species) and abundance (Brown 1984; Gaston et al. 2000). Abundance and occupancy represent distinct yet interconnected facets of population dynamics. Abundance refers to the actual count of individuals belonging to the target species within a landscape, whereas occupancy pertains to the proportion of the landscape inhabited by the target species (MacKenzie and Nichols 2004). Both these parameters have a logical link to ecological persistence, which is a determinant of long-term population dynamics. It becomes the backbone and the most relevant scale for measuring the outcomes of conservation practices or determining spaces where these practices should be applied.

Static single-species occupancy models play a crucial role in wildlife population ecology, particularly in assessing the impacts of land use change on specific species at a small scale. These models provide a valuable framework for understanding how habitat alterations influence species occurrence and persistence, offering insights into ecological processes that may otherwise go unnoticed. By focusing on the presence or absence of a single species within discrete sampling units, such as patches of habitat, static occupancy models allow the quantification of species-covariate relationships with precision. This approach facilitates the identification of key variables driving species occurrence, aiding in the development of targeted conservation strategies tailored to the needs of individual species.

In recent years, the application of single-species occupancy models has gained considerable traction in ecological research due to their versatility and robustness in assessing the impacts of anthropogenic disturbances on wildlife populations. Studies by

MacKenzie et al. (2002) and Royle and Dorazio (2008) have demonstrated the efficacy of these models in detecting subtle changes in species occupancy resulting from land use change, such as habitat fragmentation, urbanization, and agricultural expansion. By integrating spatial and temporal variability in habitat conditions, static occupancy models offer a nuanced understanding of how human activities shape species distributions at local scales. This information is invaluable for informing land management decisions aimed at mitigating the adverse effects of land use change on biodiversity and ecosystem functioning. Overall, static single-species occupancy models serve as indispensable tools for wildlife population ecologists striving to unravel the intricacies of species-environment relationships and guide conservation efforts in an increasingly human-modified world.

1.6. The White-browed bush chat.

The White-browed bushchat (*Saxicola macrorhynchus*) is a small passerine bird reminiscent of the whinchat, characterized by its relatively petite size and distinctive long, slender bill. Male specimens are adorned with striking blackish upperparts, complemented by a dark facial mask and a contrasting white chin, further accentuated by a prominent whitish eyebrow. During the breeding season, males exhibit a heightened contrast in their plumage, appearing darker and more vividly patterned compared to their nonbreeding counterparts.

In contrast, female white-browed bush chats display a more subdued appearance, featuring brownish overall plumage with pale underparts, a delicately pale eyebrow, and a lightly streaked brown back. Both male and female individuals are commonly observed foraging on the ground, engaging in a peculiar display behavior known as the "puff-and-roll," characterized by puffing out the chest and executing side-to-side movements.

This species, although rare, is primarily found inhabiting dry, arid scrublands, where it has faced population declines due to agricultural intensification and encroachment,

leading to its classification as Vulnerable. Historical records indicate its presence in low-lying arid regions of north-western India, including western Uttar Pradesh, Punjab, Haryana, south-eastern Rajasthan, and Gujarat, with recent sightings confirming its continued existence in the Thar Desert, along the western borders of Rajasthan and Gujarat, in proximity to the Pakistan border. Despite its historical distribution, the White-browed bushchat has experienced a marked decline, rendering it a single-country endemic species with a dwindling overall population. In terms of dietary preferences, the species primarily feeds on insects, including beetles, ants, and flying insects, procured either from the ground or through aerial sallies.

Information regarding breeding habits remains scarce, with limited data available on nesting behavior, clutch size, and incubation periods. While reports suggest varying patterns of movement, including sedentary behavior and localized migration, the majority of sightings occur during winter months, spanning regions from Saurashtra and Haryana to Goa, suggesting seasonal movements across its range.

2. Aims and Objectives

2.1. Objectives

2.1.1. Objectives

1. To compare and estimate the occupancy of the White-browed Bushchat across different land use patterns.
2. To estimate the Density/Abundance of the White-browed Bushchat across different land use types.

2.1.2. Questions.

1. What effect do changes in land use and land cover bring about in the occupancy of the White-browed Bushchat?

2.1.3. Prediction / Expectation: Occupancy of the White-browed Bushchat will be greater in grasslands as compared to rangelands and fallow croplands.

2.1.4. Hypothesis: Occupancy of the White-browed bush chat declines with increasing intensity of modification in landscapes.

3. Literature Review

The White-browed Bushchat *Saxicola macrorhynchus* (Stoliczka 1872) is a seldom seen and significantly understudied avian species that inhabits the desert and semi-arid environments. Historically, it was documented in regions spanning Rajasthan, Haryana, Punjab, Uttar Pradesh, and Gujarat in India, Pakistan, and parts of Afghanistan (Ali and Ripley 1987, BirdLife International 2001, 2011, Rahmani, A. R. 2012). Surveys conducted by Rahmani (1996) in 1993-1994 across Rajasthan and Gujarat rekindled scientific inquiry into this long-neglected bird, with 86 individuals recorded across eighteen surveyed sites. Despite sporadic sightings in Rajasthan, Gujarat, Haryana, and Maharashtra over the past two decades (Deshmukh 2006, Rao 2007, Singh et al. 2017, More et al. 2018), no comprehensive status survey has been undertaken since. In Pakistan, it has been declared extinct (EX) (Roberts T. 1992), while its status remains uncertain in Afghanistan (Rahmani, A. R. 2012).

(Rahmani 1996) states that rapid habitat alteration is the only perceived threat to this species. The population is declining due to extreme changes in land use patterns such as extensive agriculture, and intensifying irrigation practices to source the croplands. Fortunately, there are no records of trapping and hunting this bird according to (Rahmani 1996). Previous research on the nesting ecology of the White-browed Bushchat species is notably scarce. Ali and Ripley (1987) characterized it as localized and resident, while Roberts (1992) suggested a sedentary behavior. However, Hume (1877a) raised suspicions regarding its migratory habits, further elaborated in subsequent work (Hume 1877b), pinpointing the breeding range to Central Asia. Rahmani (1996) and BirdLife International (2001) proposed that local movements occur for breeding purposes. Harvey (2002) also indicated that some individuals might opt out of breeding in harsh environments like the Thar Desert, potentially migrating to more hospitable regions nearby. With the increasing land use changes occurring in the area, it becomes essential to scale the effects of these changes on this passerine.

Population viability is the direct or indirect consequence of the persistence of a species. This study aims to supplement a small fraction of the larger aim to understand the population trends of the White-browed bushchat in the Thar landscape.

4. Materials and methods

4.1. Study Area

4.1.1. Location

The Thar Desert, also known as the Great Indian Desert, occupies a prominent position within the hot arid zone of Northwestern India, covering an extensive area of approximately 196,000 square kilometers (refer to Figure 1). This vast desert region, predominantly situated within the boundaries of India, encompasses approximately 85% of its total area, while the remaining 15% lies within Pakistan's territory. Geographically, the Thar Desert is delineated by the Aravalli Hills in the eastern region and the Indus River in the western region (Chlachula 2021).

4.1.2. Physical features

The desert landscape is defined by its vast expanses of level to gently sloping plains, interrupted by the presence of low, barren hills and shifting sand dunes. These dunes, shaped by both aerodynamic forces and natural processes, continually change position over time. Within this arid terrain, various soil types emerge, including desert soils, red desertic soils, sierozems, and lithosols, each characterized by coarse textures and a prevalence of calcium-bearing compounds. Despite the presence of some calcium-rich layers, the soils are generally infertile and prone to severe wind erosion.

Rainfall in the region is scarce, typically averaging around 200 mm annually, with the majority occurring during the monsoon season from June to September. However, the area also experiences intermittent droughts, occurring roughly once every three years (Rao and Roy, 2012). This arid climate contrasts with the region's paleoclimate, which was more semi-arid and wetter millions of years ago. Approximately 4,000 years ago, the climate shifted dramatically, leading to weaker monsoons, increased sand deposition, and the establishment of the current arid conditions (Dhir et al., 2018).

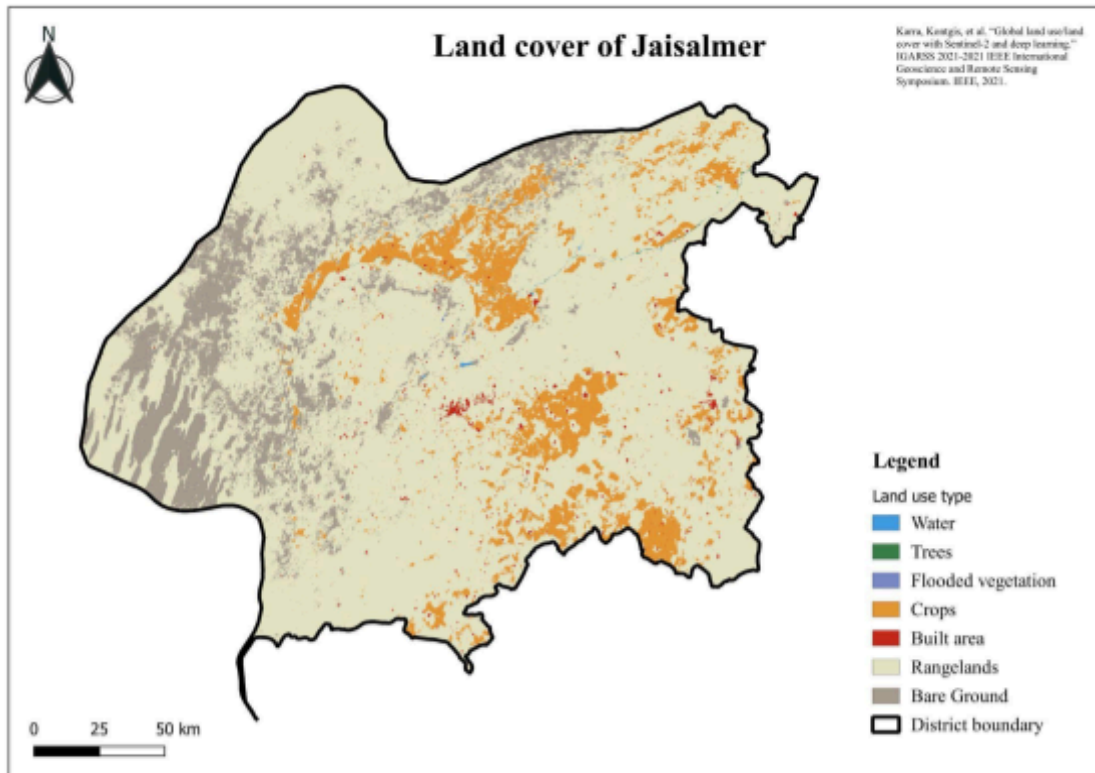


Figure 1. Landcover types of the Jaisalmer district

4.1.3. Selection of study area

The entire study area was divided into large grids of 36 square kilometers following (Dutta 2018). The study sites were chosen based on the relative habitat homogeneity of sampling grids. Although finding homogeneity in such a large area impractical, this grid was further divided into 1 square kilometer cells, ensuring coverage of all land cover types. This cell was a sampling unit for the placement of 4 line transects arranged in a square fashion.

The study was part of a larger investigation of the project titled "Monitoring threatened birds of the Thar desert". Under this project, seasonal replicates of the aforementioned line transects were taken over the last two years for the estimation of small bird populations.

4.1.4. Stratification and transect placement

Five 36 sq.km grids were selected, to estimate small bird populations. Five (± 1) representative 1sq.km. cells were randomly selected in a stratified manner. This ensured the accounting of all landscape heterogeneities across the larger 36 square-kilometer grid. Four 500m transects were placed in each of the 1 sq. km cells. Each with a distance of 150- 200m between them to ensure independence of observations. The transects which were interrupted on the ground by enclosures or agricultural fencing, were avoided to ensure ease of replicability of the transects.

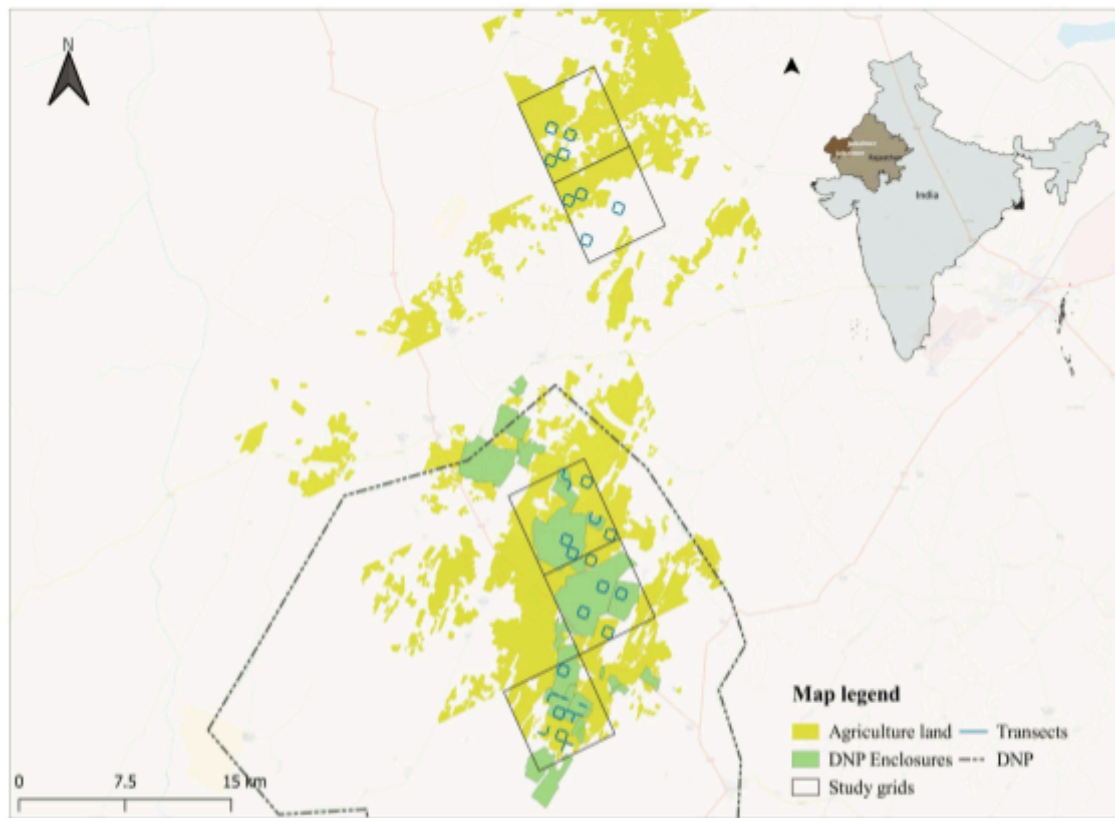


Figure 2. Study area with the sampling grids and predetermined line transects for population estimation of the White Browed bushchat

4.2. Field methods

4.2.1. Small bird surveys

During surveys conducted along 2-kilometer line transects, all perched birds were documented. Each sighting was recorded, capturing details of; species identification, the number of individuals observed, the distance of each bird from the observer, the bearing of the transect line, the angle at which the bird was observed relative to the transect line, the start and end times of each observation, and the position of the observer. All readings were recorded using the Conventional Distance Sampling protocol (Buckland et al., 2015). The positions were taken and realized on the “Locus app”, using a hand-held cellphone device. Distance measurements were recorded using a laser rangefinder (Yukon HRS300). The transect bearing and the angle of the bird for the transect were taken by a handheld Suunto compass or a compass-enabled wristwatch device.

Bird surveys were carried out from dawn until 3 to 4 hours after sunrise in the morning session and during the two hours before sunset for the evening session. These times were selected because the birds are the most active during these periods. The line transects were walked at an approximate uniform speed of 2km/h (15 to 20 minutes per 500m segment of the transect). For observing and identifying the birds in field, a binocular (Vanguard Veo ED 8*42) was used. Unidentified bird species were photographed and then identified with the help of field guides: Birds of the Indian Subcontinent. Grimmett, R., Inskipp, C., & Inskipp, T. (2016).

4.2.2. White-browed Bushchat surveys

For every small bird transect where the White Browed Bushchat was detected, additional surrounding cells were selected for adaptive cluster sampling, as the bushchat populations have been observed to be infrequent but spatially clustered, from previous surveys. Twenty-seven additional 2km transects were surveyed to get an estimate of the population density for the White-browed bushchat. The protocol remained the same except, only the bushchat detections were recorded in this particular survey.

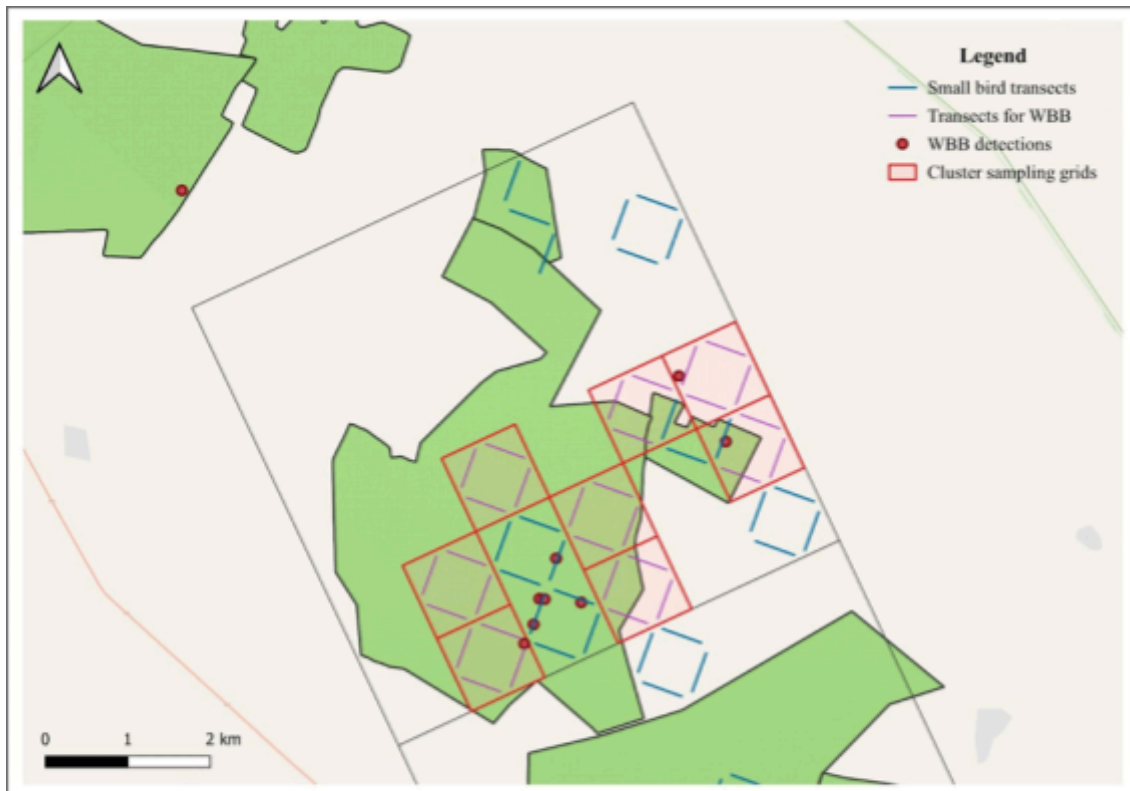


Figure 3. Illustration of the manner of selection of the adaptive cluster sampling transects, given the White-browed bushchat was detected during regular surveys.

4.3. Analytical methods

Conventional Distance Sampling (DS) was used to understand bushchat population estimates. All calculations were made according to (Buckland 2015), where the detection probability was estimated by assuming it to be a function of increasing perpendicular distance from the transect.

$$N = n/p$$

Where N is the population estimate, n is the number of individuals observed, and p is the detection probability.

$$p (\text{Probability of detection}) = \text{area under the curve} / \text{area of the rectangle}.$$

Various key functions fitted to get the best detection probability curve. All the analysis was done in R.

Single-season Occupancy Models:

Single-season occupancy models estimate the probability of site occupancy while accounting for imperfect detection. These models typically use binary data indicating

the presence or absence of the species at each survey site. Analytical methods for single-season occupancy models include maximum likelihood estimation (MLE), which estimates parameters such as occupancy probability (ψ) and detection probability (p) by maximizing the likelihood of the observed data.

5. Results

5.1. Occupancy of the White-Browed bushchat.

The naive occupancy (MacKenzie 2006) of the Bushchat in the entire study area was found to be 0.145. The naive occupancy refers to the ratio of the sites with at least one detection to all sampled sites. However, the naive occupancy does not take “imperfect detection” into consideration, thus ending up highly underreporting the occupancy of a species.

Here, occupancy models were run using the “RPresence” package. Models were run using different parameters of occupancy probability (using varying land covers as the covariates) and detection probability. The highest occupancy probability was found in Restored Grasslands (RG) which validates the hypothesis of decreasing occupancy with an increase in the intensity of modifications to the landscape. This was similar to the encounter rate found across land cover types, with significantly higher ER (0.33/ km) in Restored Grasslands followed by Croplands (0.21/ km) and Rangelands (0.18/km).

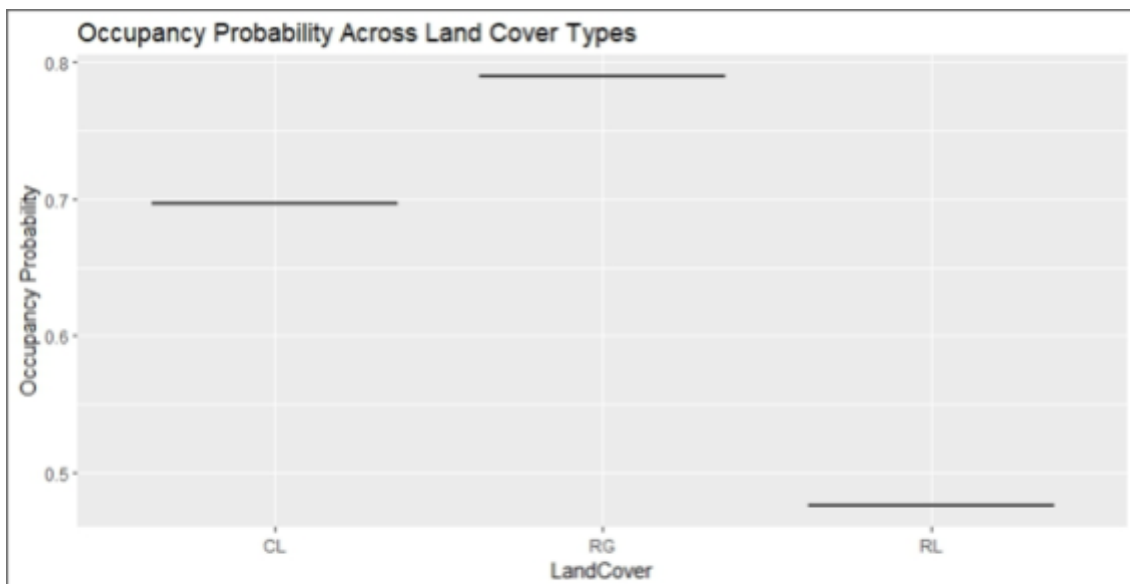


Figure 4. Occupancy of the White browed bushchat in different land cover types

5.2. Distance Sampling

5.2.1. Exploratory data analysis.

Exploratory analysis is important to get an understanding of the data, prior to fitting detection functions. It helps in deciding the truncation (w) for the detections, and explore potential patterns in potential covariates.

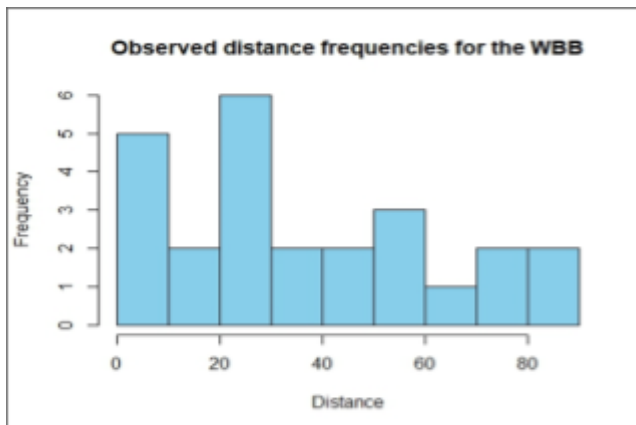


Figure 5. Frequency of distance bins

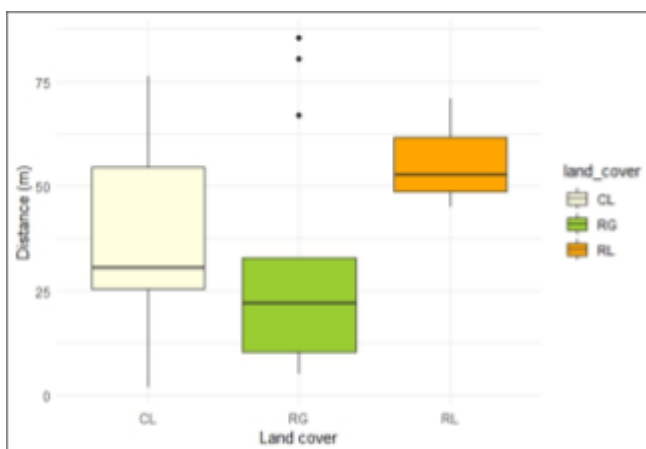


Figure 6. Boxplot of distances in each land cover type

5.2.2. Selection of appropriate detection function.

From the detection function model that best fitted our data (half normal key, adjustments= NULL), the detection function as shown in Fig. was plotted, and superimposed on the histogram showing the detection probability as a decreasing function of increasing perpendicular distance from the transect line to the detected species. Before making an inference from the detection function, goodness of fit was assessed using the `gof_ds` function from the 'Distance' package in R. According to

Buckland et al. (1993), a significant goodness-of-fit statistic is a useful warning that the model might be poor, or that an assumption might be seriously violated.

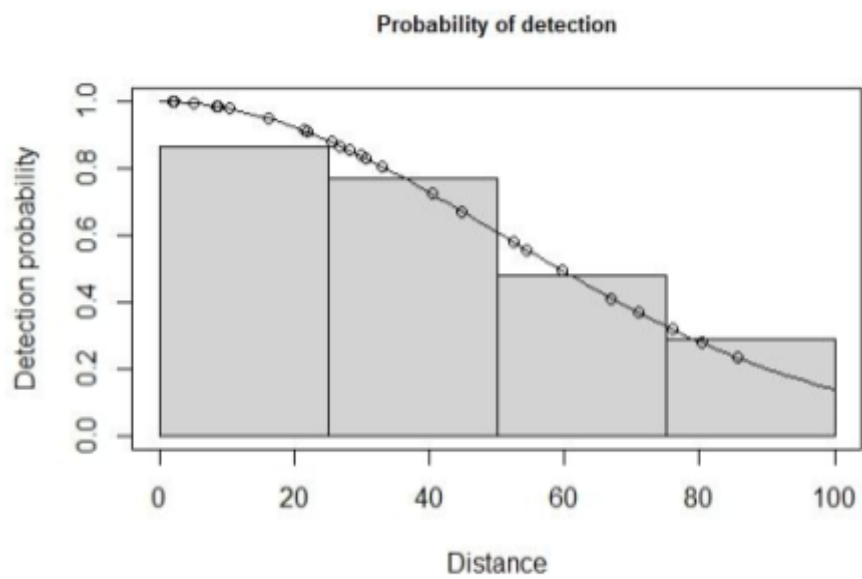


Figure 7. Histogram showing the detection probability as a function of the perpendicular distance from the transect line, truncation distance (w) set at 100m.

Summary for distance analysis:

Number of observations: 25
Distance range : 0 - 100

Model : Half-normal key function, adj = NULL (best fit)
AIC : 226.002

Estimation of Detection Probability:

	Estimate	SE	CV
Average p	0.6003329	0.1062081	0.1769154

The goodness of fit or the detection function was carried out by the Cramer-von Mises test. It evaluates how well the observed data fit the theoretical distribution assumed by the model. Distance sampling Cramer-von Mises test (unweighted)

Test statistic = 0.025793 p-value = 0.98786, this suggests a good fit.

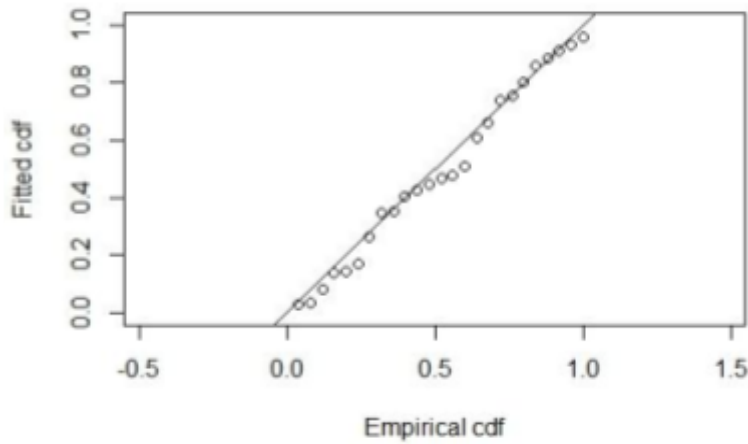


Figure 8. Cramer-von Mises test (unweighted)

The function was selected using the following values and estimates were calculated.

Model	Key function	Formula	C-vM p-value	$\hat{P}_a$	SE($\hat{P}_a$)	ΔAIC
{hn01}	Half-normal	~ 1	0.9878667	0.60033291	0.10620815	0.000000000
{hr01}	Hazard-rate	~ 1	0.2025255	0.85793122	0.00911246	0.594992624
{hn.lc}	Half-normal	$\sim \text{land_cover}$	0.8594618	0.59021453	0.11538301	3.019070288
{hr.lc}	Hazard-rate	$\sim \text{land_cover}$	0.1546433	0.88086050	0.09288608	9.023582263

5.2.3. Density/Abundance estimates.

We observed (n)25 White-browed bushchats during the winter survey period from December 2023 till February 2024, and the estimates are as follows;

1. Encounter Rate

Region	Nc	n	Effort	Covered_area	ER	ER_CV	ER_df
Thar	41.6435	25	97.77	19.554	0.25570	0.20369	195

ER: The encounter rate (ER) is 0.255702. This figure represents the average number of encounters with the white-browed bushchat per unit effort.

2. Density.

Region	n	Covered_area	Density (^D)	Density_CV	density_se
Thar	25	19.554	2.129669643	0.269794706	0.5745736

The density estimate of the White-browed bush chat in the Thar region is approximately **2.13** (± 0.57) individuals per square kilometer, with a coefficient of variation indicating moderate variability in the density estimate across observations

Density was estimated using the formula: $\hat{D} = n/a * \hat{P}_a$

Where n (number of WBBs) = 25, a (covered area) = 19.55 sq.km, and $\hat{P}_a$ (probability of detection) = 0.6003329.

3. Abundance

	Estimate	SE	CV
N in covered region	41.6435602	9.0555114	0.2174529

For the study area, the abundance estimation was found to be 41.64 (± 9.05) individuals.

Abundance was estimated using the formula: $N = \hat{D} * a$

Where N (Abundance) = 41.6 individuals, $\hat{D}$ (Density) = 2.13 individuals/sq.km, and a (Area covered) = 19.55 sq.km.

6. Discussion

6.1. Population estimation and occupancy of the White-browed bush chat

The results of the study suggest that the occupancy of this infrequent species is affected by changes in land uses like extensive agriculture and overgrazing of pastoral lands. This study represents a novel investigation of the model species with no previous history of any systematic study. Hence this study might potentially provide a framework to monitor infrequent grassland species. These methodologies could be explored in other landscapes with similar landscape compositions. The estimates suggest a very low density (2.1 individuals for every square kilometer) value for the bush chat. The study also presents differential values of occupancy with associated land cover types. This however forms a very broad metric for a species level occupancy.

6.2. Conservation Implications and Way Forward.

Significant changes in the encounter rates suggest the need to mitigate the challenges of extensive land use changes in the Thar landscape.

Interestingly, cropland areas demonstrated a greater value for occupancy probability for the White-browed bushchat. The observations made in this land cover type were usually on the fringes or periphery of the agricultural lands, where there was a prevalence of shrubs and moderate-height grasses. These areas were generally located beside open grassland patches. Intensively grazed rangelands showed the least occupancy probability for the bushchat.

One individual was ringed in the RKVY enclosure inside the Desert National Park. This marked the first global record of the passerine being captured and ringed. This could potentially give insights into the unknown, and highly contested realm of migration for this species.

6.3. Caveats of the study

The very obvious caveat for the study was the inability to use continuous habitat data as covariates for estimating occupancy. Quantification and measurements of classified upper-story and under-story vegetation would make the occupancy models more robust. This could also be used to extrapolate the occupancy parameter to associated habitat covariates, to get the potential sites of occupancy for the White-browed bushchat.

The adaptive cluster sampling utilized for infrequent and clustered species can cause us to overestimate abundance, due to an increase in the sampling effort in the areas of greater animal abundance. Due to the short duration of the study, all these parameters could not be taken into consideration.

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